Paper Review: An Analysis of Facebook Photo Caching (Huang et al., SOSP 2013)

Facebook serves photos through four layers: the browser cache, flash-based Edge Caches located at points of presence, an Origin Cache that spans four US data centers and is addressed by hashing the photo ID, and the Haystack storage backend. The Edge and the Origin both use FIFO eviction, and each resized version of a photo is cached as its own object. The paper asks how much traffic each layer absorbs, how requests move geographically, whether other cache sizes or eviction policies would perform better, and whether metadata such as photo age or the owner's follower count predicts access. To answer these questions, the authors instrumented desktop browsers, Edge hosts and Origin hosts for one month. Sampling was based on a hash of the photo ID, so a sampled photo appears in the trace of every layer and its events can be matched across layers. The resulting trace contains 77.2M requests from 13.2M browsers and is also replayed in trace-driven cache simulations.

The paper's main contribution is the first cross-layer measurement of a production image-serving stack, from the client down to storage. Browser caches serve 65.5% of requests, the Edge 20.0%, the Origin 4.6% and Haystack the remaining 9.9%. Popularity follows a Zipf distribution near the client and becomes flatter at each deeper layer. Photos that many users each view once ("viral" content, in the authors' terms) benefit little from browser caching and are served mostly by the shared Edge. Edge assignment weighs latency, capacity and peering cost, so many requests are served far from the user, and 17.5% of clients use more than one Edge, which adds cold misses. In simulation, S4LRU improves on FIFO's object hit ratio by 8.5% at the Edge and 13.9% at the Origin, and a single collaborative Edge running S4LRU reduces Origin-to-Edge bandwidth by 42%. The authors also find that request rates decline with photo age in a roughly Pareto pattern and that follower count predicts popularity only for public pages.

A trace of this scale and realism is difficult to obtain outside a large provider, and sampling by photo ID is a sound choice because it retains unpopular photos and allows events to be matched across layers. The authors also check their sample for bias. The study nevertheless has several limitations. Its coverage excludes mobile apps, Akamai-served traffic and non-US PoPs, and the trace spans a single month, so the high browser hit ratio may not generalize. Browser hits are inferred rather than observed, by matching browser and Edge requests on URL and IP address, and NATs, proxies and changing IP addresses can distort that matching. In the bias check, hit ratios shifted by as much as 4.3%, and the reported Edge hit ratio is about 5% lower than Facebook's live monitoring. The paper gives no error bars, although some of the claimed gains (LFU's +2%, for example) are smaller than this variation. In the simulations, the current cache size is estimated by fitting the FIFO curve to the observed hit ratio, and the paper does not explain how cache sizes in a sampled trace translate to real capacity. Because the Clairvoyant baseline ignores object size, it is not an upper bound on byte-hit ratio. Hit ratio is also the only performance measure. User-visible latency is not measured, and the paper does not discuss the extra flash writes and metadata that S4LRU would require, which matters because FIFO suits flash well. The latency and peering costs of a collaborative Edge are mentioned but never quantified. Finally, the policy comparison is limited to FIFO, LRU, LFU and S4LRU and omits size-aware policies such as GDSF and adaptive ones such as ARC and LIRS.

A natural extension would be to trace mobile clients, Akamai paths and international PoPs over a longer period, and to report confidence intervals for every inferred hit ratio. Size-aware and age-aware eviction policies could be compared with S4LRU in a simulation that accounts for flash write amplification. The case for a collaborative Edge would be stronger if its hit-ratio gains were weighed against a cost model of cross-country latency and peering charges. The simulated improvements should also be validated with a controlled A/B deployment on live traffic.